



UniMelb
Information Security Club

Careers in Cyber

Exploring professional pathways, roles, and hands-on skill development

Blue Team / SOC

Red Team / OffSec

Security Engineering / Cloud



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About Me

Damien Lowe – Education Director

My Experience

INCOMING ROLE

Junior Security Analyst @Asahi

PAST EXPERIENCE

Security Analyst Intern @Fortian

INTERESTS

Broad cyber security interests, currently interested in **physical** systems.



EDUCATION

2nd Year Bachelor of
Science

Major in Computing & Software
Systems

Diploma in Languages
Japanese



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The SOC Analyst BLUE TEAM

Monitor logs. Triage threats.

As a Security Operations Center (SOC) analyst, you are the first line of defense. Your primary responsibility is continuous monitoring and rapid response to maintain security posture.

Analyse system logs and network traffic for anomalies

Investigate and triage security alerts in real-time

Contain threats and coordinate incident response

*If you want this job, you need to understand **cloud infrastructure**.*

That is exactly why we are running key hands-on workshops this semester:

WEEK 8

Azure Cloud Security Workshop

WEEK 10

AD 101 (Active Directory)



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Offensive Security Red Team

Diverse Domains of OffSec

Offensive security is a mindset and a toolkit for identifying and proactively closing security holes. We explore various disciplines to build a complete defensive understanding.

Infrastructure Security

Assess vulnerabilities in networks, active directory, and cloud environments.

Application Security

Examine compiled and source code for memory corruption, logic, and flaws beyond just web apps.

Hardware & IoT

Dissect and exploit connected devices, smart systems, and custom hardware.

Social Engineering & Red Teaming

Understand psychological manipulation and simulate full-scope adversary attacks.

Vulnerability Research

Discover previously unknown flaws (zero-days) and build reliable exploits.

Your Pathway in OffSec

Our workshops are designed to give you foundational skills and hands-on experience across multiple domains.

FOUNDATIONAL MODULES (CORE SKILLS)

- Linux Networking & Core System Administration
- Operating System Internals (Windows & Linux)

SPECIALIZED DOMAIN MODULES

- **Web Application Security:** Burp Suite, Wireshark, and modern web flaws.
- Infrastructure Pentesting & Active Directory
- Cloud Security (AWS, Azure)
- Introduction to Assembly & Reverse Engineering

ADVANCED CHALLENGES & PROJECTS

- Exploit Development
- Organized Red Team Exercises
- Homelab of simulated enterprise environment



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Governance, Risk & Compliance (GRC)

The Reality of GRC

Everyone wants to be a hacker, but GRC is where the actual enterprise budget lives.

You map business risk to security controls. If you understand the technical side and the business side, you write your own ticket.

Compliance Frameworks

ISO 27001 International Information Security Standard

Essential Series Australian Cyber Security Centre Mitigation Strategies



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Security Engineering **SecEng**

Designing, building, and automating secure systems to stop threats.

What They Actually Do

- **Focus:** Building & Hardening
- **System Architecture:** Designing with modern tools
- **Threat Modeling:** Continuous proactive defense

Essential Skills & Tech

- **Scripting & Automation:** Python, Bash, CI/CD
- **Infrastructure:** Cloud Security & Orchestration
- **Security Tooling:** Implementation of SIEM/EDR solutions

Typical Day-in-the-Life

- **Code Reviews:** Auditing source and cloud configurations
- **Automation:** Engineering custom security workflows
- **Hardening:** Applying patches and strict systems hardening

Career Pathways & Entry

- **Feeder Roles:** SysAdmin, Software Developer, NOC
- **Entry Roles:** Security Operations, Jr Security Engineer
- **Key Certs:** CompTIA Security+, AWS Certified Security, BTLI

We are running an Industry event later this semester where you can meet people in all of these roles

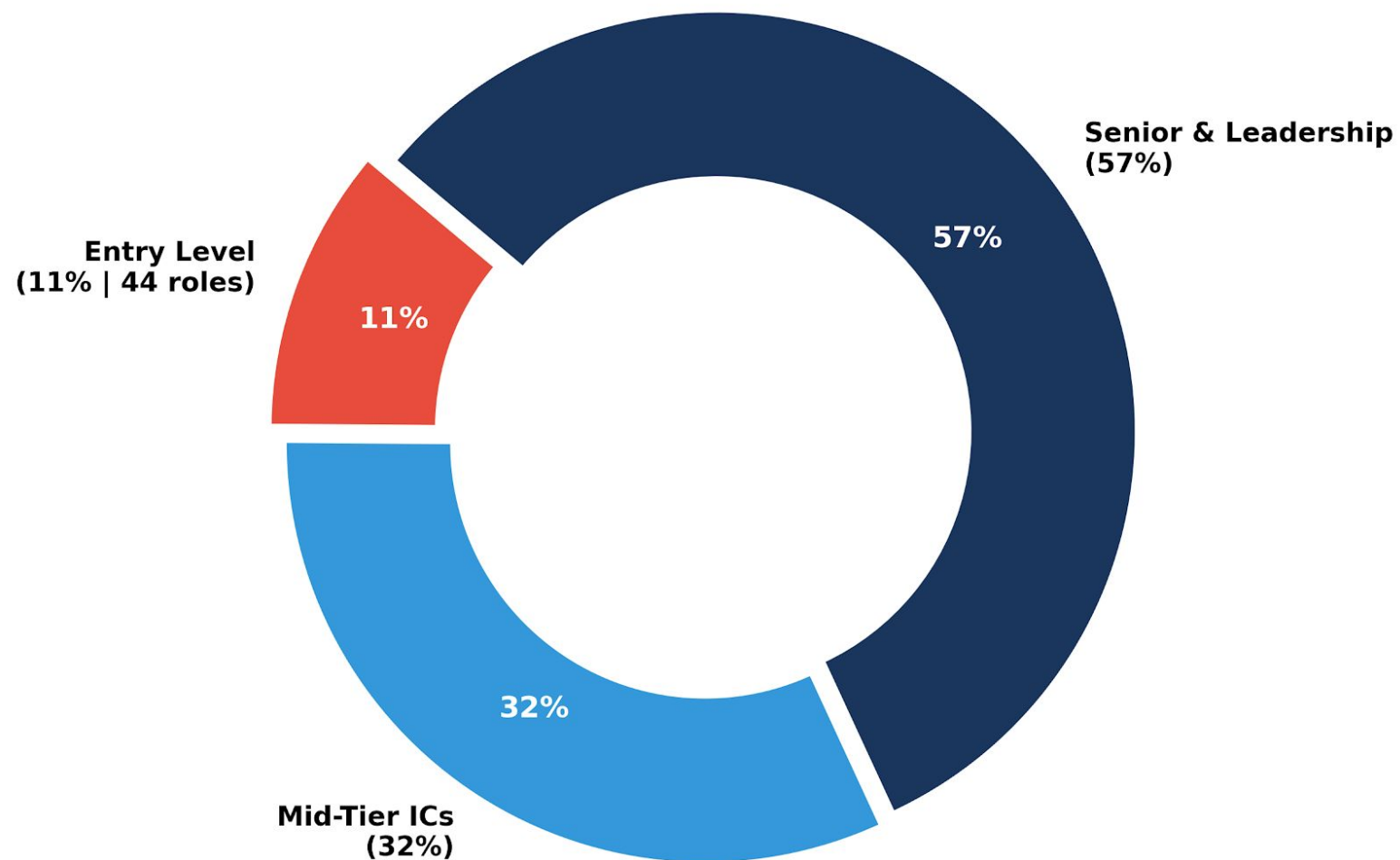


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The Entry-Level Myth: The Experience Filter

The job market isn't broken, it's just competitive for beginners.



This is the **scariest chart** you will see all year.

Only **11%** of open cyber security roles in Australia are entry-level.

That represents exactly **44 jobs** across the entire country.

The market does not want to train you. They demand immediate utility, heavily prioritizing **Senior & Leadership talent (57%)**.

*Take this with a grain of salt it's a relatively small sample size but gives a good idea of the industry

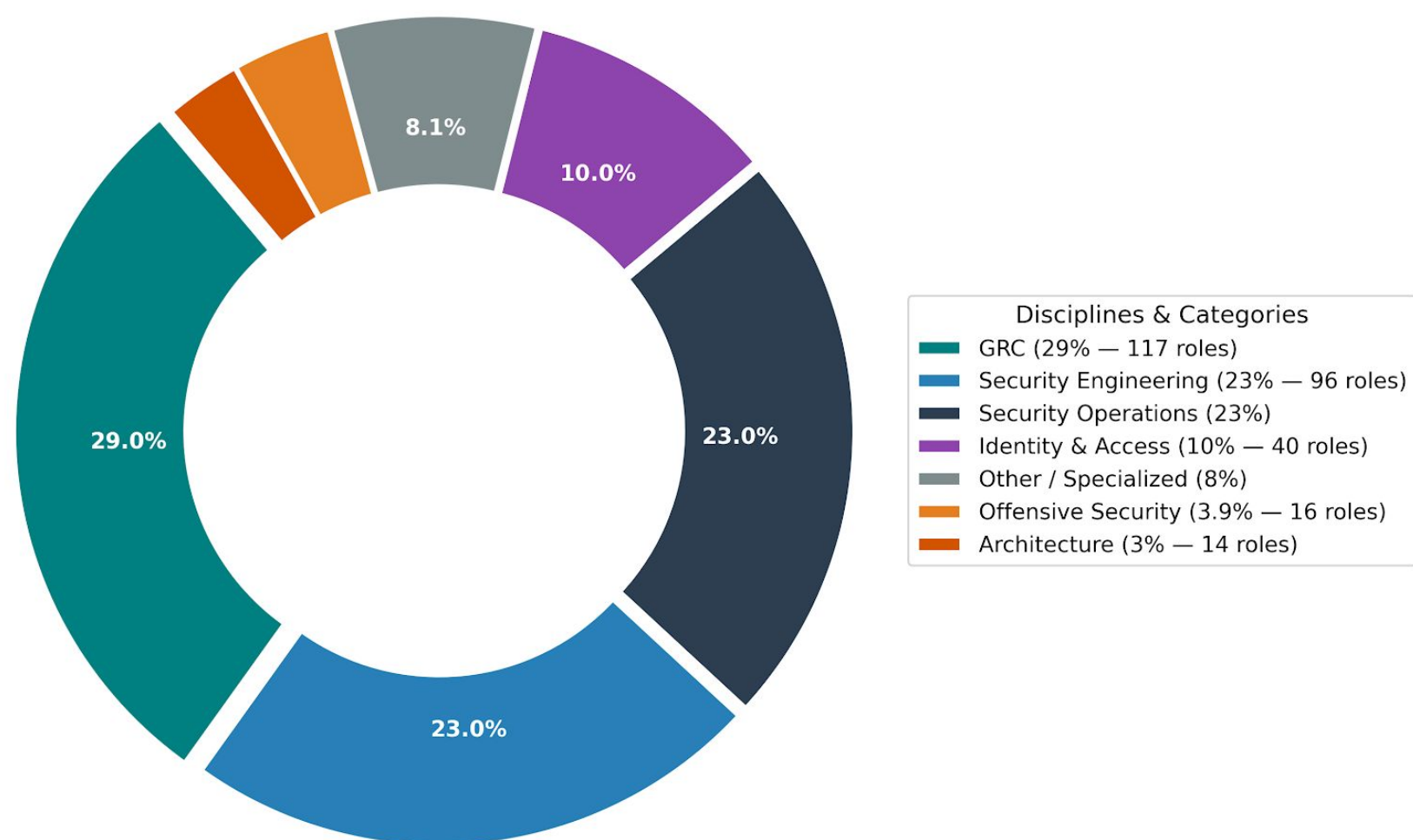


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The Industry Delusion: Stop Fighting for Scraps

Australian Cyber Security Jobs by Category / Discipline
June 2026



Source: CyberSec People Jobs Report (June 2026) | Base: 409 In-Scope Roles

Everyone wants to be a pentester. But the market has other plans.

Offensive Security makes up exactly **3.9% of the Australian job market**—only 16 active roles.

Meanwhile, **GRC represents 29%** and **Security Engineering commands 23%**. If you only learn to break things, you are limiting your opportunities. Build infrastructure and map compliance to secure your place.

OFFENSIVE ROLES

3.9%

16 Roles Nationally

DEFENSIVE & GRC

52.0%

213 Roles Nationally



The Visa Filter **vs.** The Competence Filter

Structural hurdles are keeping you out, not your lack of potential.

The Automated Barrier

"Lazy HR Filters": Instantly automated barrier

- **Automated PR/Citizenship checkboxes** filter out non-residents first
- **ATS instantly rejects** profiles lacking legal working status
- **Talent discarded** before a human recruiter ever sees the resume

The Employer Motivation

What's Really Happening: The logic behind the filters

- **Visa processing** represents a heavy \$5k–\$15k direct business expense
- **HR risk aversion** and legal friction deter international sponsorship
- **"Safety" is chosen over skill** when engineering timelines are tight

Waste of energy

Defense & Public Sector (AGSVA Clearances): Closed for non-citizens right now



The Bypass: How to Become Undeniable

Stop playing the HR portal game. Start building proof of work.

THE GOLDEN RULE OF THE BYPASS

Standard portals filter you out. The people you meet bring you in.

Private Enterprise

Target Strategy

- **SaaS & Cloud-Native:** High-growth organizations with agile hiring.
- **Major Banks & FinTech:** Complex systems hungry for secure talent.
- **Tech Consultancies:** Always seeking proven, hands-on engineers.

Public Artifacts

Proof of Work

- **Active GitHub Repos:** Publish tools, custom scripts, and integrations.
- **HackerOne / Bugcrowd:** Verifiable profiles with real-world impact.
- **Technical Write-ups:** Deep-dives explaining how you broke or built systems.

Economic Leverage

High-Value Skills

- **Automating SOC Logs:** Prove you can eliminate tedious manual triaging.
- **Terraform & IaC:** Deploy hardened infrastructure mathematically.
- **Saving Team Headcount:** Build automations that multiply engineering output.



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Companies Hiring International Students

Scan to access our curated directory of visa-friendly employers.

01

80+ Companies

Curated list of companies actively supporting international talent.

02

Job Search Boards

Target roles that bypass automated citizenship filters.

03

Resume Builder/Grader

Find sites that build and grade resumes.



SCAN THE QR CODE

Access the employer directory instantly



The HR Bypass: **Blue Team**

Build a live SIEM & honeypot deployment to create undeniable proof of engineering competency.

01

Deploy Cloud SIEM

Spin up Azure Sentinel (Free Tier) to act as your centralized security operations center.

Expose Honeypot

Deploy a vulnerable Windows VM honeypot and deliberately expose RDP (Port 3389) to the public internet to attract active global brute-force attacks.

02

Ingest Log Streams

Configure Windows Event Viewer to feed failed logon events directly into your Sentinel workspace.

Geographic Mapping

Write custom Kusto Query Language (KQL) scripts to extract IP addresses, correlate them with geo-data, and map live brute-force login attempts worldwide.

03

Document on GitHub

Do not just do the work, prove it. Create a public repository containing your architecture diagrams, active configuration files, and custom KQL scripts.

The Artifact Portfolio

Embed dynamic dashboards and geographic threat maps in your README. Share this repository on LinkedIn and embed it directly on your CV to bypass automated HR filters completely.



The HR Bypass: Red Team

Stop running automated tools. Build them. Write custom exploits to prove mechanical understanding.

01

Develop Custom Exploits

Find a recent, simple CVE from the last 12 months. Write a custom Python script to automate the detection and exploitation of that specific vulnerability rather than relying on black-box scanner tools.

02

Publish & Document on GitHub

Responsibly publish the script to GitHub with a full technical write-up breaking down the memory mechanics or weblogic. **Hiring managers want to see that you understand the underlying code.**

exploit_scanner.py

GitHub Repository • Active Source Code

```
# Custom exploit scanner for CVE-2025-XXXX
import socket, sys
def check_vulnerability(target_ip, port):
    payload = b"\x90\x90\x90\x41\x41\x41\x90"
    try:
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.connect((target_ip, port))
        s.send(payload)
        resp = s.recv(1024)
        if b"VULN_ECHO" in resp:
            print(f"[+] Host {target_ip} is VULNERABLE!")
        else:
            print(f"[-] Host is patched.")
    except Exception:
        print(f"[!] Connection failed.")
```




Bypassing the Portal

How to bypass automated filters and deliver your project portfolio directly to decision makers.

01 / THE PORTAL TRAP

Portal Applications Guarantee Failure

Applying through online applicant tracking systems (Workday, Seek) instantly screens out qualified candidates based on automated keyword and questionnaire filters.

02 / THE RECON

Target the Decision Maker

Identify the SOC Manager, Head of Security, or Lead Engineer at a tech company on LinkedIn. Avoid HR, target the person who actually runs the pipeline.

03 / THE PAYLOAD

Send a Review Script

Do not ask for a job. Ask for professional critique on your project. This completely bypasses early-stage bureaucratic HR filters.

Direct Message

Active Now • LinkedIn Mobile

H

Thanks for connecting. Let me know if you ever want to run through your background.

Hi [Name],

I'm a UniMelb student. I just built a custom Azure Sentinel honeypot using Terraform and documented the brute-force attack logs.

Would love to get your eyes on the architecture and portfolio.

Write a message...



The Physical Override

If the HR portal filters you out, you have to stand in front of the person who controls the budget.

01 / THE BASELINE

MISC Workshops

Learn the technical mechanics. Before you step out into the industry network, build your baseline competency inside our weekly club workshops.

02 / THE MONTHLY CADENCE

Meetups

Attend consistently. Do not ask for internships, buy a security engineer a beer and ask about their day-to-day life in the role

03 / THE GOLDEN RULE

Do Not Hand Out Resumes

Hand out GitHub links instead. Prove your technical worth with live repository builds, custom exploit scripts, and well-documented cloud architecture diagrams.

Direct Networking Playbook

The Mechanical Hack for Industry Placement

The people who can override a strict resume screen or an experience filter do not hang out on university campuses. They are at BSides, SecTalks, and OWASP meetups.

Make it a rule: attend one industry meetup every month. The network you build at these events is a guaranteed thing that will improve your chances at getting hired

MAJOR NODES & TARGET CONFERENCES

BSides

Ruxcon

AppSec

LTOTM

Hck4G



Careers & Pathways Q&A

Ask your questions before we jump into the practical tabletop exercises.

01 / ASK ANYTHING

Discussion Prompts

- **Portfolios:** What projects grab the most attention from SOC Managers?
- **Meetups:** Which local networking events are most welcoming for newcomers?
- **Workshops:** How to balance university study with extracurricular cyber builds?
- **Interviews:** What non-technical soft skills do security teams look for?
- **Interviews:** How can I prep for an interview? (We can help!!!)

Next Steps

Help us improve • Plan your semester



Scan for Feedback

- 📄 (W2) Career - Homelab
- 📄 fortian presentation
- 📄 W8 cloud security | azure
- 📄 W10 - Intro to AD 101
- 📄 W5 - Burpsuite
- 📄 W4 - Workshop (Endpoint)
- 📄 W7 - CTF Workshop pt 2
- 📄 W7 - CTF Workshop pt1
- 📄 Industry Panel x Edu Wkshop
- 📄 W5 - (Wireshark)
- 📄 W1 - Intro to MISC



SOC Analyst Tabletop

01 / DETECT & TRIAGE

Analyze the Noise

Your shift starts at the alert queue. As a SOC Analyst, you must quickly differentiate false positives from critical security incidents, extracting indicators of compromise (IoCs) like malicious URLs, IPs, and file hashes before they cause harm.

02 / CONTAIN & PROTECT

Stop the Spread

When a real threat is confirmed, speed is key. You will execute urgent containment protocols: disabling compromised credentials, revoking active session tokens, isolating infected hosts from the network, and blocking adversary command infrastructure.

03 / INVESTIGATE & PREVENT

Piece the Puzzle

Immediate containment is just the beginning. You will reconstruct the threat actor's timeline. How did they breach the network? What did they access? Your final analysis forms the playbook to prevent the next intrusion.



Stage 01: The Phishing Report

01 / THE INCIDENT DETECTED

TIME: 2:15 PM

THE ALERT: A user from the Finance department forwards an email to security@company.com.

SUBJECT: "URGENT: Q3 Payroll Policy Update."

"Hi team, this looks suspicious. Is this safe?"

– Finance Department User

What is your very first move as a SOC Analyst?

Interactive Tabletop Playbook

Analyst Discussion • Immediate Triage Action

[] **Extract Email Headers** — Parse hops, SPF/DKIM validation, and origin IP.

[] **Verify Sender Domain** — Inspect for typosquatting or character spoofing.

[] **Harvest URLs & Hashes** — Safely extract links/attachments for sandbox/VT analysis.



Stage 02: **The Click**

Tabletop Scenario: Walk through the first critical moments of an active incident trigger.

02 / THE ESCALATION

TIME: 2:20 PM

THE REALITY: Users lie. You check the web proxy logs. The user didn't just forward the email; they forwarded it *after* they already clicked the link.

THE ARTIFACT: The link redirects to a microsoft login page @login-microsoft365-update.com.

The link was clicked. What is your immediate action?

Interactive Tabletop Playbook

Analyst Discussion • Immediate Triage Action

- **[] Password Reset** — Force an enterprise password reset immediately.
- **[] Session Revocation** — Revoke all active session tokens for the user's account.
- **[] Azure AD Audit** — Check Azure AD logs for anomalous logins or MFA fatigue activity.



Stage 03: **The Payload**

Tabletop Scenario: Walk through the first critical moments of an active incident trigger.

03 / THE EXPLOIT RUNNING

TIME: 2:25 PM

THE ALERT: Your Endpoint Detection & Response (EDR) platform flashes red.

THE REALITY: The user didn't just type in their password. The website downloaded a file named Q3_Payro11.pdf.exe. The user bypassed the Windows warning and ran it as an Administrator.

Malware is actively executing on a Finance endpoint. What do you do to the machine?

Interactive Tabletop Playbook

Analyst Discussion • Immediate Triage Action

- **[] Network Isolate Host** — Quarantine the machine via EDR immediately.
- **[] Do NOT Power Down** — Keep host alive; shutting down loses critical forensic RAM.
- **[] Pull File Hash** — Extract executable signature for IOC generation and search.



Stage 04: The Breach

Tabletop Scenario: Walk through the first critical moments of an active incident trigger.

04 / THE BREACH & ESCALATION

TIME: 2:30 PM

THE TELEMETRY: Before you could isolate the host, the .exe spawned a hidden PowerShell window.

THE ACTIONS: It established an outbound connection to an unknown Russian IP (Command & Control). It then started scanning port 445 (SMB) across your entire internal network.

The attacker has a reverse shell and is moving laterally. What is the play?

Interactive Tabletop Playbook

Analyst Discussion • Immediate Triage Action

- **Outbound C2 Beacons** — Check firewall logs for connection patterns to the external IP.
- **Active Directory Audit** — Inspect domain controllers for newly created admin accounts.
- **Incident Response Plan** — Formally initiate containment, eradication, and recovery phases.



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Thank You!

We'd love your feedback.

Your insights help us shape workshops to better suit you.

Please scan the QR code on the right to complete a quick feedback survey and share your experience from today's session.

FEEDBACK PORTAL

Scan with your mobile device to respond

